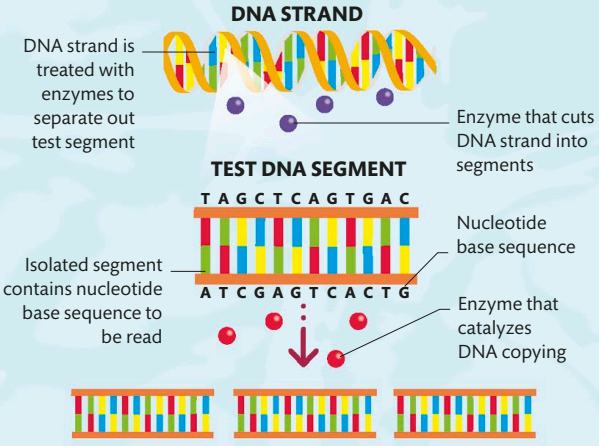
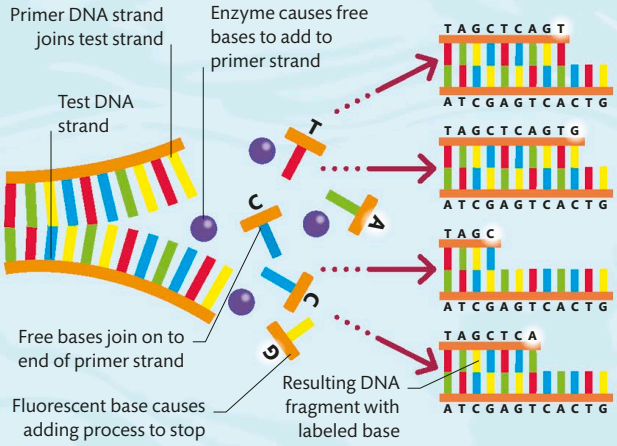
Gene testing

Some tests allow scientists to detect abnormalities in individual genes, such as extra or missing material, or bases in the wrong place. The samples are examined by a method such as DNA sequencing, which reveals the order of nucleotides in a section of DNA. The presence of an abnormality does not always indicate a problem; it may be a variation that has no ill effects. However, some abnormalities can affect health, so expert interpretation of test results is important.

DNA sequencing
In one widely used method of DNA sequencing, modified fluorescent nucleotide bases (see opposite) are added to the ends of DNA strands to highlight each base in a DNA strand. There are four types of fluorescent tags—one for each type of nucleotide base (A, T, C, or G).



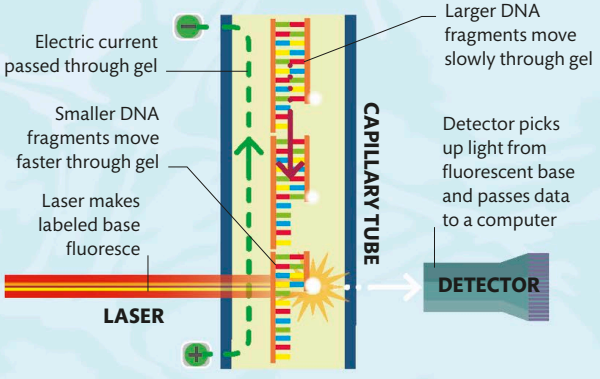


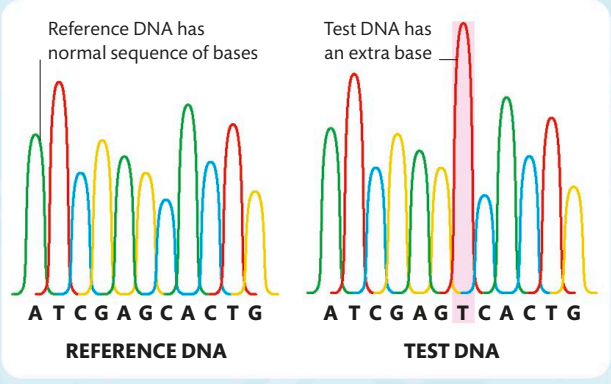
1 Isolating test DNA segment

A DNA sample may be obtained from various sources, such as cheek cells, saliva, hair, or blood. The sample is treated with an enzyme that cuts the DNA into segments in order to isolate the segment of DNA to be analyzed. Using another enzyme, this test DNA segment is then copied hundreds of times to produce a sample that is large enough for analysis.

2 Labeling bases in the test DNA

The test DNA sample is mixed with "primer" DNA, an enzyme, free nucleotide bases, and nucleotide bases labeled with a fluorescent marker. The primer joins onto the test strand, and the free bases join onto the ends of the primer. This process stops when a fluorescent base is added. Each resulting DNA fragment ends up with a labeled base corresponding to one base on the test DNA.





3 Detecting labeled bases in the test DNA

The DNA fragments are passed through gel in a thin tube (capillary). An electric current makes the fragments move, and they end up sorted by length; the order of the labeled bases reflects the order of bases on the test strand. As each fragment passes the laser, its labeled base fluoresces, and the detector reads each of these in order.

4 Computer analysis

The detector passes the sequence of bases in the test sample to a computer. The computer uses the data to produce an image called a chromatogram, in which the nucleotide sequence is shown as an image and as letters. The test DNA chromatogram is compared to one of a normal reference DNA sample to identify any differences.